

Accounting for variability in the truth-evaluation of epistemic possibility statements

Giuseppe Ricciardi & Joshua Martin

Harvard University - Meaning & Modality Linguistics Laboratory

01/06/2022

Epistemic modals and perspective

Two types of epistemic statements depending on whether they are overtly anchored to a body of information (a perspective)

- ❑ **Overtly anchored: relativized epistemic statement (*relativized EP*)**

E.g., “Given what I know, John might be home now”

- ❑ **Not overtly anchored: bare epistemic statement (*bare EP*)**

E.g., “John might be home now”

An implicit perspective in bare EPs

- ❑ Consensus among semanticists

bare EPs are covertly relativized EPs

$[[\text{modal } p]]^w = [[\text{given the evidence } E, \text{modal } p]]^w = [[\exists / \forall w': w' \text{ compatible with } E \text{ in } w. p \text{ is true in } w']]^w$

- ❑ Disagreement on **what count as the relevant perspective when evaluating the truth-value of a bare EP**

(e.g., Kratzer 1981, 1986, 2012; MacFarlane 2011, 2014; Egan et al 2005; Egan 2007; Yalcin 2007, 2011; Stephenson 2007; von Stechow & Gillies 2008, 2011; Egan & Weatherston [ed.] 2011; Bach 2011; Stalnaker 2014; Knobe & Yalcin 2014; Yanovich 2014; Khoo 2015; Moss 2015; Swanson 2015; Dowell 2017; Beddor & Egan 2018; Khoo & Phillips 2019; Phillips & Mandelkern 2020; Rudin 2021)

Main goal of this talk

To argue for these claims:

- ❑ Utterances of bare declarative EPs are covertly relativized to a perspective as much as utterances of any other declarative sentence are
- ❑ Perspective-taking does not play a more special role in the truth-evaluation of an epistemically modalized statement than in the nonmodalized cases

Crucial data in this debate

Truth-evaluation of bare “might p” in scenarios where

- ❑ **“might p” is assertable based on the speaker’s evidence**
- ❑ **“might p” is not assertable based on the truth-value assessor’s evidence**

Crucial data in this debate

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- ❑ **“might p” is assertable based on the speaker’s evidence**
- ❑ **“might p” is not assertable based on the truth-value assessor’s evidence**

E.g.,

A says: “John might be home now”

You know that

1. the event of John being home now is compatible with A’s body of information
2. John is not home now

Would you judge what A said as true or false?

A call for experimental work

- ❑ Let's ask folks out there

- ❑ Previous experimental works in this direction:

Knobe & Yalcin (2014); Beddor & Egan (2018); Khoo & Phillips (2019);
Phillips & Mandelkern (2020)

Preview

- ❏ Knobe & Yalcin 2014

- ❏ Exp 1

- ❏ Exp 2

- ❏ Exp 3

- ❏ General discussion

Knobe & Yalcin 2014, Experiment 4: methodology

Sentence (between-subjects)

Sally and George are talking about whether Joe is in Boston. Sally carefully considers all the information she has available and concludes that there is no way to know for sure.

Sally says: “Joe is in Boston” (**Nonmodal**) / “Joe might be in Boston” (**Modal**)

Just then, George gets an email from Joe. The email says that Joe is in Berkeley. So George says: “No, he isn’t in Boston. He is in Berkeley”

Test question (Between-subjects)

Retraction

We want to know whether it would be appropriate for Sally to take back what she said (for example, by saying “Ok, scratch that”). So please tell us whether you agree or disagree with the following statement:

- It would be appropriate for Sally to take back what she said.

Falsity

We want to know whether what Sally said is false. So please tell us whether you agree or disagree with the following statement:

- What Sally said is false.

Task

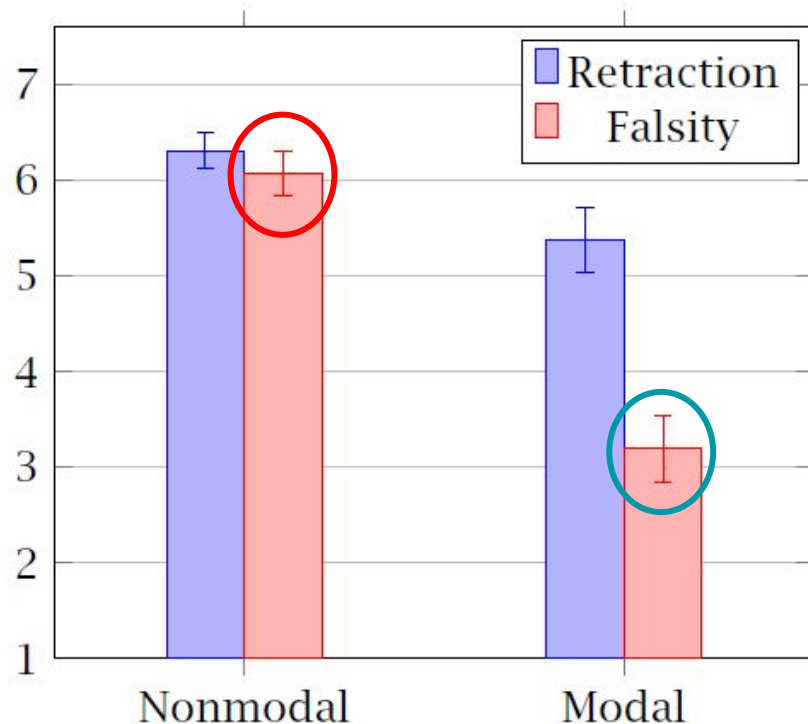
All participants were asked to indicate their level of agreement on a scale from 1 (“completely disagree”) to 7 (“completely agree”).

Knobe & Yalcin 2014: Exp 4

Two results highlighted:

1. The mean falsity rating of modal is in the middle of the scale (at about 3)
1. Mean falsity rating of the nonmodal high in the scale (at about 6)

Both results are replicated in Phillips & Mandelkern (2020)



Overview of our studies

- ❑ Exp 1: investigate what is underlying finding 1 and whether it was due to task effects
- ❑ Exp 2: investigate whether finding 1 extend to the statements “it is possible that p” and “may p”
- ❑ Exp 3: investigate what is underlying finding 2

Experiment 1: methodology

Knobe & Yalcin's task modified:

- ☐ Critical statement underlined (cf. Doran et al. 2012)
- ☐ Explicit question about the assertability of the statement by Sally
- ☐ Simple truth-evaluation question (simpler question than the original) with a forced choice between “true” and “false”

Please read the following scenario, and then answer the questions below.

Sally and George are talking about whether Joe is in Boston. Sally carefully considers all the information she has available and concludes that there is no way to know for sure.

Sally says: 'Joe might be in Boston'



Just then, George gets an email from Joe. The email says that Joe is in Berkeley. So George says: 'No, he isn't in Boston. He is in Berkeley.'

KY_Justification

Based on what she knows, is Sally justified to say the underlined statement?



Justified



Not justified



KY_Truth

Is the underlined statement true or false?



True



False



Experiment 1: methodology

Knobe & Yalcin's story modified:

Original

Sally and George are talking about whether Joe is in Boston. Sally carefully considers all the information she has available and concludes that there is no way to know for sure.

Sally says: 'Joe might be in Boston'

Just then, George gets an email from Joe. The email says that Joe is in Berkeley. So George says: 'No, he isn't in Boston. He is in Berkeley.'

- a) Participants are not told any positive evidence for Sally's claim
- b) participants have to assume George's perspective
- c) There is **short time gap** between when George hears the statement (the utterance time) and when he receives the news that Joe is not in Boston

New

Imagine that you are sitting in a café in Berkeley with a friend of yours named Sally, having a conversation about another friend, Joe.

Sally says: "Joe might be in Boston, since he had a business meeting scheduled there this week."

You know that Joe's meeting in Boston was cancelled and he is in Berkeley right now.

- a) Sally's evidence supporting her 'might p' claim is made explicit by Sally herself (**overt relativization**)
- b) Sally is talking directly to participants
- c) Participants know that Joe is not in Boston at the utterance time

Experiment 1: methodology

Design 2X2X2

Factor 1: Story	("Original"; "New")	manipulated between-subjects
Factor 2: Target judgement	("Justification"; "Truth")	
Factor 3: # of judgements per participant	("Between" [only one]; "Within" [both with justification preceding truth])	

Participants: recruited 50 Mturk workers per condition

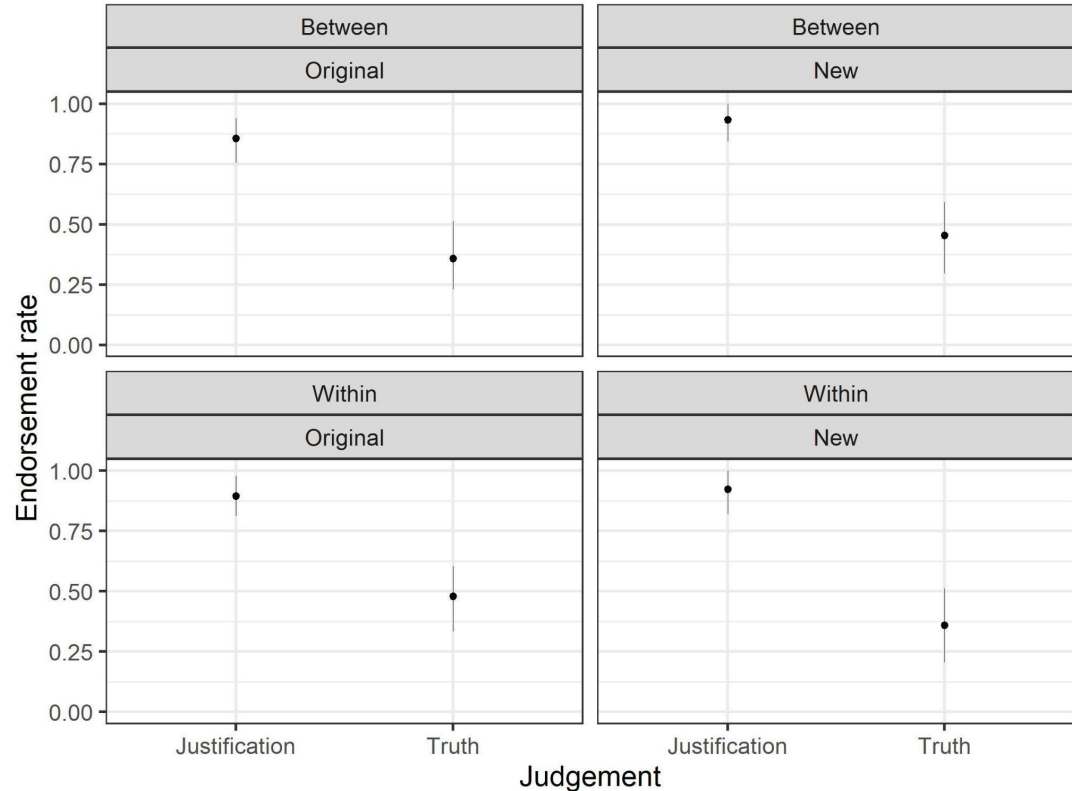
Exclusion criteria: participants had to be native speakers of English, from US, answer correctly the question: "What was the name of the woman talking in the story?"

Experiment date: 06/10/2021

Experiment 1: results

Results:

- ❑ convergence in the Justification condition
- ❑ split in the Truth Condition



Experiment 2: methodology

Design (2 conditions)

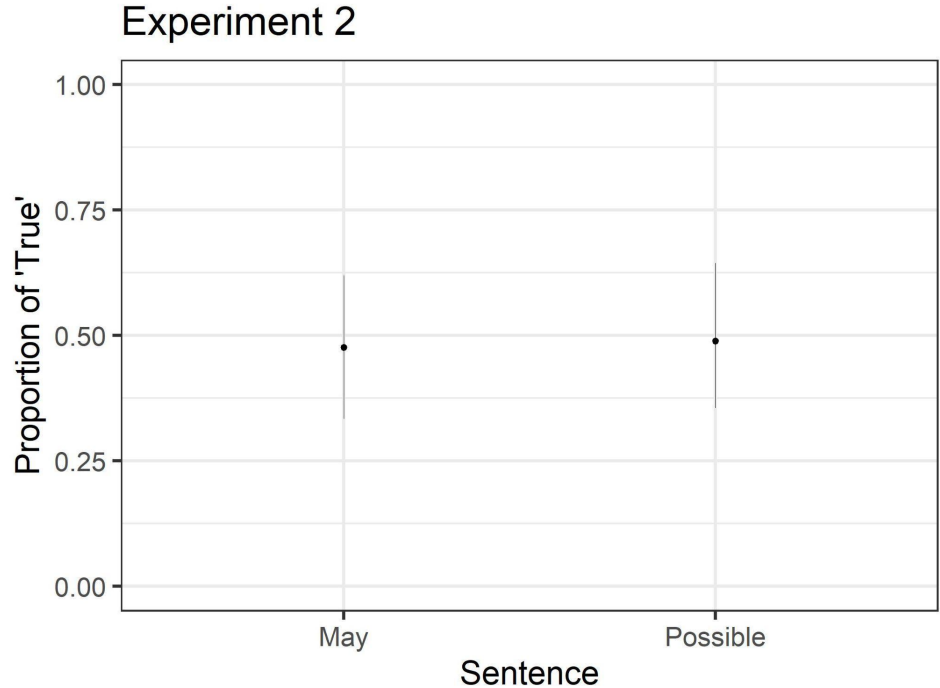
- ❑ Same task as Exp 1;
- ❑ only original story;
- ❑ only truth judgment;
- ❑ Factor: Sentence type (“Possible”; “May”) manipulated between-subjects

‘It is possible that Joe is in Boston’ [“Possible”] / “Joe may be in Boston” [“May”]

Experiment 2: results

Results:

Same finding as for “might p”:
even split among participants in
the truth-evaluation of “may p”
and “it is possible that p”



Experiment 1 and 2: discussion

The finding:

Split among participants in the truth-evaluation of Sally's utterance of bare "might/may/possible [Joe is in Boston]" when Sally is justified to say so but participants know that Joe is not in Boston

Why?

Experiment 1 and 2: discussion

Answer 1: Split in the population of English speakers about the interpretation of the truth question when asked about epistemic modal statements

An epistemic modal statement “does not serve to describe the world. There is no way the world could be, or could fail to be, which would settle the question of the truth of the sentence [...] When ordinary speakers are asked, ‘Is what [the speaker] said true?’ Some of them interpret the question as [someone equipped with the information as the speaker would rationally make the utterance?], and they answer ‘yes’. Others interpret the question as [someone equipped with the full information would rationally make the utterance?], and they answer ‘no’”

(Yalcin 2011: 311-312) ¹⁹

Experiment 1 and 2: discussion

Answer 2: Split in the population of English speakers about the interpretation of the truth question

- ❑ For some (those who answered “true”), the question is interpreted as “true/false from the speaker’s perspective with limited info”
- ❑ For some, (those who answered “false”), the question is interpreted as “true/false from the perspective defined by the full information”

Cf. Reuter & Brun (2021): correspondence vs coherence views of truth

Experiment 1 and 2: discussion

The finding:

Even split among participants in the truth-evaluation of Sally's utterance of bare "might/may/possible [Joe is in Boston]" when **Sally is justified to say so but participants know that Joe is not in Boston**

Why?

- ❑ Answer 1: Split in the population of English speakers about the interpretation of the truth question only for epistemic statements
- ❑ Answer 2: Split in the population of English speakers about the interpretation of the truth question

Experiment 3: rationale

To discriminate between the two answers, it is crucial to look at the behavior with “bare p”:

- ❑ If a similar split is **not** found with “bare p”, then Answer 1 is right
- ❑ If a similar split is found with “bare p”, then Answer 2 is right

Experiment 3: no conflict in the original nonmodal

Sally and George are talking about whether Joe is in Boston. Sally carefully considers all the information she has available and concludes that there is no way to know for sure.

Sally says: Joe is in Boston

Just then, George gets an email from Joe. The email says that Joe is in Berkeley. So George says: 'No, he isn't in Boston. He is in Berkeley.'

Is Sally justified to make the underlined statement based on what she knows?

If not, the case of “bare p” is not parallel to the case of “might p”: there is no conflict of perspectives

Experiment 3: trying to make Sally's statement assertable

It's late afternoon and Sally and George are in a pub in Berkeley talking about a common friend, Joe. Yesterday, Joe told Sally that he had a job interview in Boston at 5pm today and he would fly there early in the morning.

So, Sally says: "Joe is in Boston"

Just then, George gets an email from Joe. The email says that the job interview was cancelled and that he is still in Berkeley. So George says: "No, he isn't in Boston. He is in Berkeley."

Experiment 3: methodology

Design 2X2

Factor 1: Sally's justification status (“Justifiedp”; “Nonjustifiedp”)

manipulated between-subjects

Factor 2: Target judgment (“Justification”; “Truth”)

manipulated within-subjects with justification preceding truth

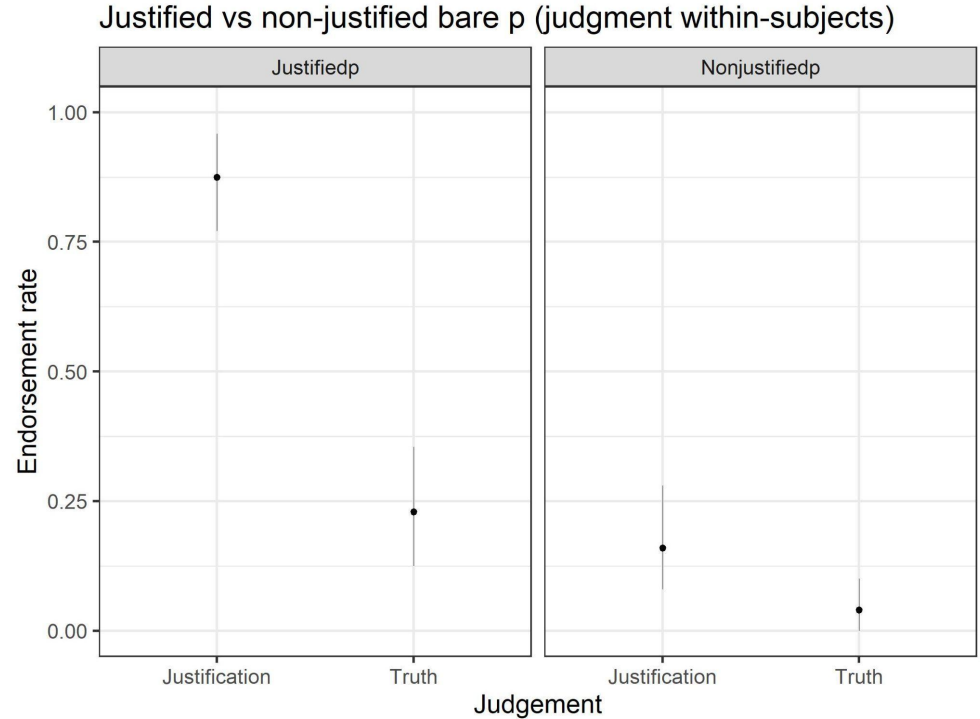
Participants: recruited 50 Prolific workers per condition

Exclusion criteria: same as experiment 1

Experiment 3: results

Results:

- Results analyzed with a logistic regression model with “Sally’s justification status” as effects-coded fixed effect
- Main effect of Sally’s justification status such that **the proportion of participants who answered “true” was significantly higher in the Justified condition than in the Non-justified condition** ($\beta = -1.9650$, $z = -2.459$, $p = 0.0139$)
- Planning to run a study with more stories

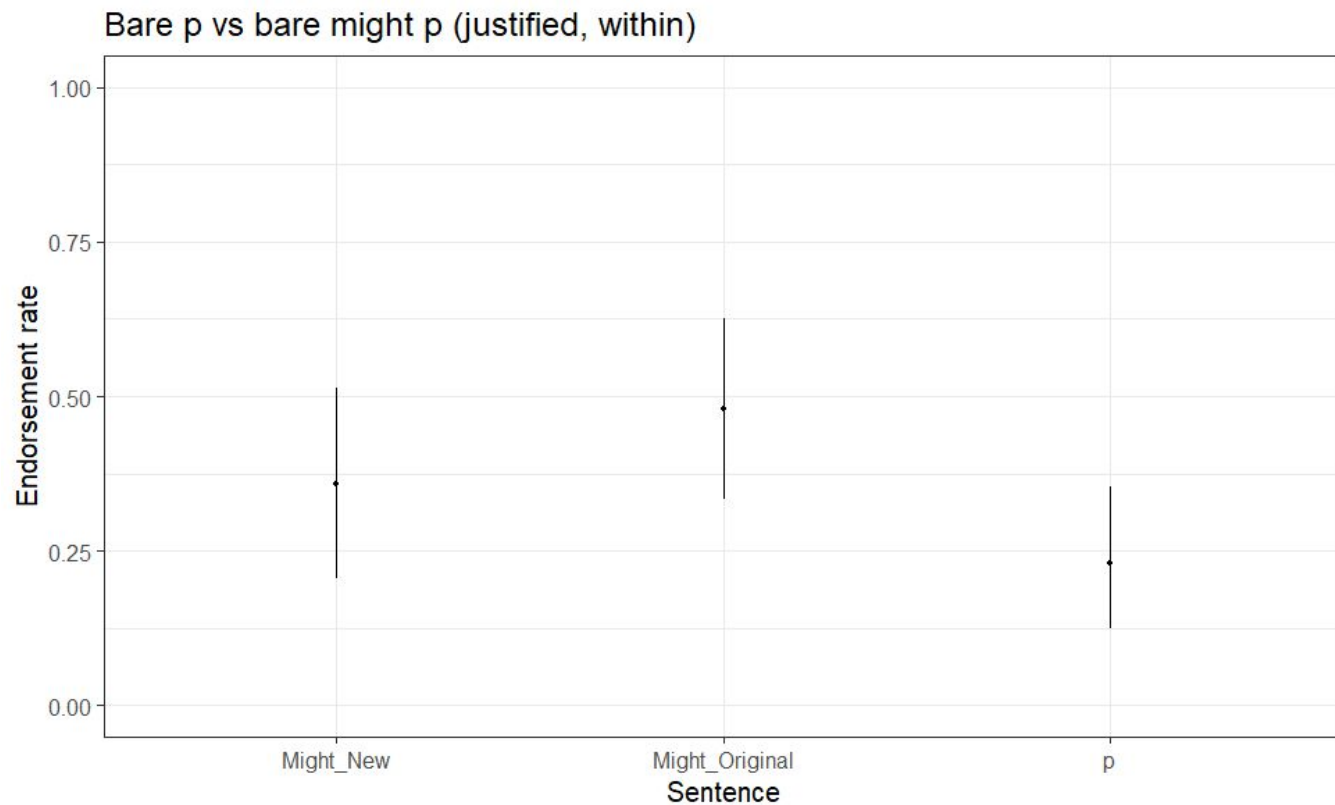


Experiment 3: discussion

- ❑ To account for the behavior of participants in the “Justified p” condition of experiment 3 it’s plausible to assume that:
 - Those who answered “true” assumed Sally’s perspective
 - Those who answered “false” assumed their perspective

- ❑ Cf. Reuter & Brun (2021) for even more striking results with similar stories presenting conflicting perspectives: participants evenly split in their truth-evaluation

General discussion



General discussion

- ❑ Participants' behavior in truth-value assessing a statement coherent with the speaker's evidence but not with theirs' doesn't qualitatively differ between "bare p" and "bare might p":
participants split between the choice "true" and the choice "false"
- ❑ We have good reasons to explain this behavior by postulating a split in the perspective adopted for the truth-evaluation

What does this behavior explained in this way suggest about the semantics of "bare p" and "bare might p"?

General discussion: the theory

Based on insights from Bach (2011), MacFarlane (2011), Harris (2020), and Rudin (2021)

- ❑ “The semantic value of an expression is a constraint on what speakers can say with the expression, and not the content of what they say” (Harris 2020: 1). That is, semantic value = character in Kaplan’s sense
- ❑ Every sentential semantic value comes with an indexical information state variable i_x defining what the facts are for an agent x (**epistemic statements are no special**)

- Semantic value (=character) of p = $\lambda w.[p \text{ is true in } \mathbf{w}_{ix}]$
- Semantic value (=character) of *might* p = $\lambda w.[\exists w': w' \text{ compatible with } \mathbf{w}_{ix} . p \text{ is true in } w']$

$\mathbf{w}_{ix}$ = the world w as seen from the perspective of the information state i held by the agent x

- ❑ Acts of uttering of a sentence s come with an assignment of value to the variable i_x (cf. Rudin 2021: 3 “When a speaker assert a sentence [...] they are predicating something of their epistemic state”)
- ❑ Truth-value assessments of a sentence s also come with an assignment of value to the variable i_x which **doesn’t need to coincide with the assignment defined in the original context of utterance**

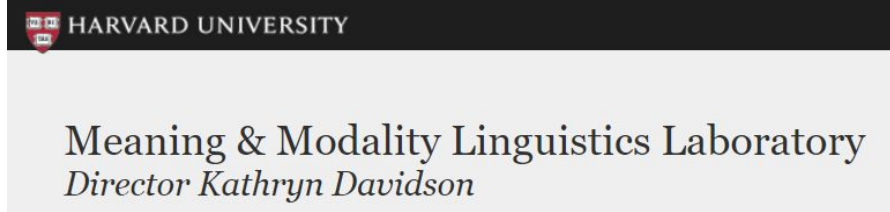
Conclusions

- ❑ We have offered findings suggesting that the truth-value assessment of an uttered declarative sentence where the speaker's perspective conflicts with the assessor's perspective is sensitive to this difference, no matter whether the sentence is epistemically modalized or not
- ❑ Specifically, participant treat the utterer perspective in evaluating the truth of a sentence in the same fashion across the modalized / nonmodalized divide:
some include it, some exclude it
- ❑ In other words, whether the utterer perspective is considered as part of the “message conveyed by a sentence utterance” is not a specific problem of epistemic statements
- ❑ **These findings cast doubt on the hypothesis that perspective-taking has a more essential relation to the semantics of an epistemic sentence than to that of a non-epistemic sentence**

Open questions

- ❑ How does this perspective-sensitivity of the truth-value assessment relate to our understanding of the meaning of the assessed sentences?
- ❑ **What does truth-evaluation reveal us about the semantic value of declarative sentences?**

Acknowledgements



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